

A SMART PACKAGING PLATFORM For Safe, Secure Delivery



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Now is the age of online shopping and home delivery. In this context, we often hear stories about things not going the way they ought to. A customer replaced original products with dummy ones and returned those to a renowned online dealer, making huge profits. A person thrashed a company on social media for failed delivery, only to realise that the product had been claimed wrongly. Products get stolen during delivery, also exposing private information to unwanted eyes.

These, while not everyday occurrences, are serious problems, especially for the vendor. But, the system is here to stay. What can then be done to ensure security of every delivery that is planned? Let us put together a smart packaging platform, says the team from Nexshark, emphasising the importance of vendor-customer handshake.

Say Hi to OpeKun

Adithya Venkatesh, founder and CEO of Nexshark, and his team consisting of Sivakumar, Nitin Kumar, Uma Sankar and Prakash, went about designing an innovative platform to link e-commerce vendors and end users in an electromechanical way, to ensure safe delivery of packages.

In e-commerce delivery, there is no

authentication or authorisation between the consumer and the retailer. Named OpeKun, this solution makes use of Bluetooth Low Energy capabilities and uses strong encryption methods to acknowledge safe receipt of parcels.

How it works

The idea is to create a strong handshake between the vendor and the customer. Package boxes are designed with a combination of electronic and mechanical methods.

Basic principles like operation of levers, pulleys and gears, 3D printing, mobile computing and Bluetooth connectivity have been exploited to arrive at this comprehensive low-power solution. Each package has a microcontroller with miniaturised semi-conductors for reduced size and a lock that is connected to the Cloud; the entire process is encrypted by asymmetric keys.

When a parcel is dispatched, a public key is programmed into the microcontroller inside the package. At the same time, a private key is securely delivered to the customer on the app. The public key is embedded in iBeacon format and transmitted; though everyone can see it, it is ineffective unless combined with the private key.

Once the parcel arrives, the customer

Fig. 1: Innovative packaging

